Sahaj Bhadja

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Summary

Platform Software Engineer with 2+ years of experience building **multi-cloud automation**, **secure access controls**, and reliable **containerized platforms**.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Operating Systems, Analysis of Algorithms, Database Systems

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Operating Systems, Object-Oriented Programming, Software Engineering

Experience

USC Dornsife **September 2025 – Present**
Software Engineer *Los Angeles, CA*

- Architected a **multi-tenant Azure platform** supporting 500+ users with row-level isolation, normalized schemas, Blob Storage, and indexed production access patterns, reducing dashboard latency from 1.2s to 350ms.
- Achieved 99%+ reliability across 5K+ monthly jobs through **fault-tolerant automation**, validation, reconciliation, error isolation, cloud storage, observable state transitions, incremental processing, and stable repeatable CI/CD-backed production releases.

Grad8 **June 2025 – August 2025**
Software Engineer Intern *Los Angeles, CA*

- Reduced feedback turnaround by 60% through a **secure Python platform** supporting 150+ instructors and 10K+ daily requests with REST APIs, AWS RDS, authenticated ingestion, and GitHub Actions.
- Processed 1K+ files per batch using **Python multithreading**, validating untrusted PDF, DOCX, and ZIP inputs while isolating failures and preserving responsive developer-facing workflows under production load reliably.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer - (SecureSync) *India*

- Increased pipeline efficiency by 68% through **AWS Glue and S3 automation**, transforming structured data while reducing storage costs by 15% and supporting repeatable cloud delivery workflows at scale.
- Automated builds and 20+ unit/integration suites through **Jenkins and GitHub**, adding security scans, failure reports, cross-platform validation across Windows, Linux, and macOS environments, and automated nightly reporting.

Aasma Technology Solutions **May 2023 – August 2023**
Full Stack Developer Intern - (VibrantChem) *India*

- Synchronized 2M+ records through an eight-service platform with dedicated **authentication and RBAC services**, JWT controls, WebSockets, Azure SQL encryption, firewalls, and secure access workflows across distributed locations.
- Reduced resource wastage by 42% through **secure delivery automation**, integrating Docker, Jenkins, dependency scanning, automated tests, Azure Monitor, Sentry, and Kibana across modular production services and delivery workflows.

Pandit Deendayal Energy University **January 2023 – May 2024**
Research Fellowship *India*

- Designed a **distributed authentication architecture** for users, IoT devices, and role-based access, evaluating smart-contract verification, immutable identity records, and self-sovereign credentials to reduce centralized identity security risks.
- Published peer-reviewed research analyzing **identity attack protections** against replay, phishing, man-in-the-middle, and credential theft, documenting authorization trade-offs and presenting defensible access-control designs internationally with peers at IEEE conferences.

Skills

Languages: Bash, Python, Java, SQL, JavaScript, TypeScript, C++
Cloud & Platform: AWS, Azure, GCP, Docker, Kubernetes, Cloud Storage, Multi-Tenancy, Distributed Systems
Identity & Security: IAM, Authentication, Authorization, RBAC, JWT, Access Control, RLS, Data Validation
Automation & Delivery: CI/CD, GitHub Actions, Jenkins, Git, PyTest, Mockito, Unit Testing, Integration Testing
APIs & Observability: FastAPI, Flask, REST APIs, Microservices, Fault Isolation, Audit Logging, Sentry, Azure Monitor, Troubleshooting

Projects

Multi-Cloud Access Provisioner - Design Concept

- Proposed a **Bash/Python access automation** service that validates role requests, applies scoped AWS, Azure, and GCP permissions, records approvals, rotates credentials, and supports reversible provisioning workflows safely.

Access Drift Auditor - Design Concept

- Proposed a **multi-cloud governance pipeline** comparing permission snapshots, flagging excessive privileges and orphaned identities, generating audit evidence, and routing prioritized remediation alerts through observable, testable workflows reliably.

Blockchain Authentication Research | Paper

- Evaluated **identity and access management** architectures for users, IoT devices, and role-based authorization, analyzing immutable records, smart-contract verification, self-sovereign identity, and secure credential sharing in peer-reviewed research.